ADC 2020 – Questions and SOLUTIONS

1. a) Using nodal analysis, calculate the voltages at nodes *X* and *Y* of Figure 1.1. [6]

KCL at X: $\frac{X-13}{1}+\frac{X-5}{2}+\frac{X-Y}{5}=0 \;\Rightarrow\; 17X-2Y-155=0$

KCL at Y: $\frac{Y-X}{5}+\frac{Y}{1}+3=0 \;\Rightarrow\; 6Y-X+15=0$

Solving the equations on right gives X = 9, Y = –1.

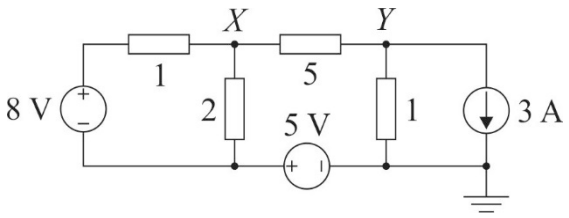


Figure 1.1

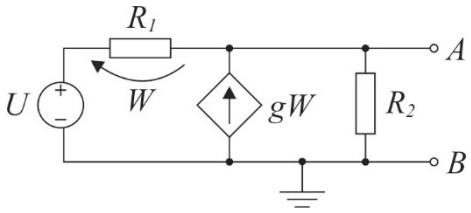


Figure 1.2

b) Draw the Thévenin equivalent of the network shown in Figure 1.2 and derive expressions for the values of its components. Note that the network contains a voltage-controlled current source. [6]

Suggested method: apply current source *I* between *A* and *B* and then derive expression for output voltage $V = V_{AB} = V_{Th} + R_{Th}I$.

KCL at *A*: $\frac{V}{R_2}-g(U-V)-\frac{U-V}{R_1}-I=0$

$\Rightarrow\; V(R_1+gR_1R_2+R_2)-(1+gR_1)R_2U-IR_1R_2=0 \;\Rightarrow\; V=\frac{(1+gR_1)R_2U+R_1R_2I}{R_1+gR_1R_2+R_2}$

So $R_{Th}=\frac{R_1R_2}{R_1+gR_1R_2+R_2}=R_2 \parallel \frac{R_1}{1+gR_1}$; $V_{Th}=\frac{(1+gR_1)R_2U}{R_1+gR_1R_2+R_2}=\frac{R_2}{R_2+R_1/(1+gR_1)}U$

c) For the circuit in Figure 1.3, derive an expression relating *Z* to *X* and *Y*. You should assume the opamp is ideal. [5]

This is an inverting summing amplifier referenced to 3 V. The gains from *X* to *Z* and *Y* to *Z* are –1/2 and –1/3 respectively. The output voltage is therefore:

$$Z=3-\frac{X-3}{2}-\frac{Y-3}{3}=\frac{11}{2}-\frac{X}{2}-\frac{Y}{3}$$

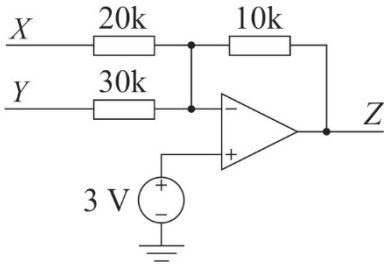


Figure 1.3

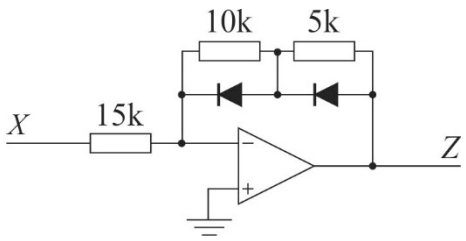
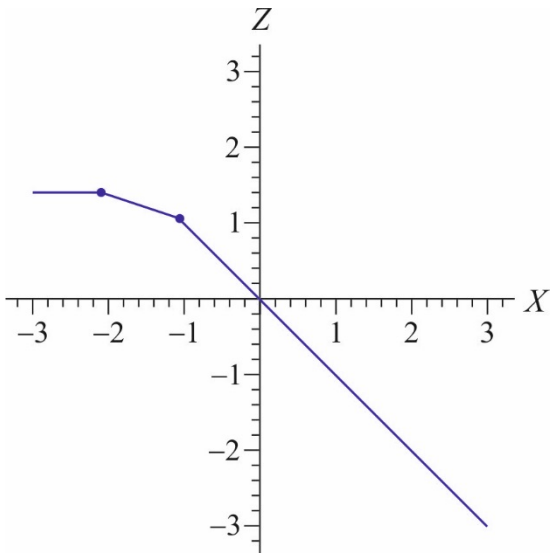


Figure 1.4

d) In the circuit of Figure 1.4, the diodes have a forward voltage of 0.7 V when conducting but are otherwise ideal. Draw a dimensioned sketch showing the variation of Z with X for input voltages in the range $-3 \le X \le 3$.

[6]

The LH diode will come into conduction when the voltage in the 10k resistor reaches 0.7 V. This occurs when $X = -\left(\frac{15}{10}\right) \times 0.7 = -1.05$ V. The output voltage at this point will be Z = +1.05 V. For X > –1.05 V the circuit is an inverting amplifier with gain –1.
The RH diode will come into conduction when the voltage across the 5k resistor reaches 0.7 V. This occurs when $X = -\left(\frac{15}{5}\right) \times 0.7 = -2.1$ V. The output voltage at this point will be $Z = 2 \times 0.7 = 1.4$ V. For X < –2.1 V the output voltage will be clamped at +1.4 V.



e) For the circuit in Figure 1.5, choose the value of R_B to give a bias voltage of 2.5 V at the collector of the transistor. [5]

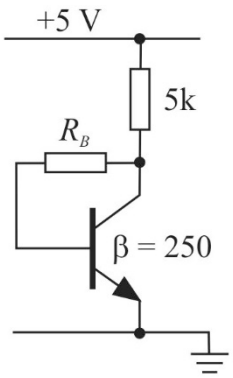


Figure 1.5

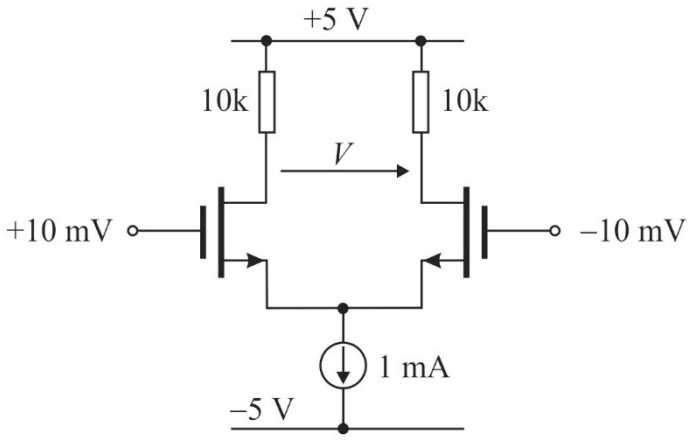


Figure 1.6

If the collector voltage is +2.5 V, the emitter current must be $\frac{5-2.5}{5k} = 0.5$ mA. The base current is then $\frac{0.5m}{1+\beta} = 1.992\ \mu A = (2.5-0.7)/R_B \Rightarrow R_B = 904\ k\Omega$.

f) The MOSFETs in Figure 1.6 are matched, with $K = 2$ mA/V^2. Calculate the differential output voltage, V, assuming the MOSFETs have infinite output resistance. [6]

Can either solve large signal equations or use small-signal approximation. Suggest latter approach as easier.
Each MOSFET is biased at 0.5 mA, so the transconductance is $g_m = 2\sqrt{KI_D} = 2$ mA/V.
The differential input voltage is 20 mV and the double-ended gain is $g_m R_D = 20$, so the differential output voltage is $V = 20m \times 20 = 400$ mV.

g) For the partial circuit in Figure 1.7, assuming the drain of the MOSFET is held at a potential such that the device is in saturation, show that the drain current is 0.1 mA. Also calculate the small-signal output resistance under these conditions. [6]

The gate voltage is zero and the source voltage is $V_S = I_D R_S$, so the drain current equation is:

$I_D = K(-I_D R_S - V_t)^2 \Rightarrow R_S^2 I_D^2 + \left(2R_S V_t - \frac{1}{K}\right) I_D + V_t^2 = 0$

$25I_D^2 - 12.5I_D + 1 = 0 \Rightarrow I_D = 0.1$ mA ~~or 0.4 mA~~. Second solution can be rejected as it would imply $V_{GS} = -2$ V which is sub-threshold.

The output resistance of a MOSFET in common source configuration with source degeneration is $R_o = r_o + R_S + g_m r_o R_S$. The transconductance is $g_m = 2\sqrt{KI_D} = 0.4$ mA/V and $r_o = \frac{V_A}{I_D} = 1\ M\Omega$, so $R_o = 1M + 5k + 0.4m \times 5k \times 1M = 3.005\ M\Omega$.

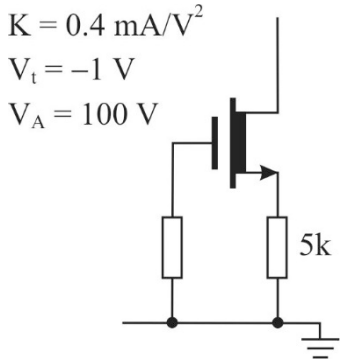


Figure 1.7

2. Figure 2.1 shows a simple power amplifier which is to be used to drive a loudspeaker. The amplifier comprises an inverting op-amp circuit with a transistor output stage to boost the output power capability.

   a) If the opamp stage is required to have an input impedance of 1 kΩ and a voltage gain of –20 in the mid-band, choose values for R_1 and R_2.

[6]

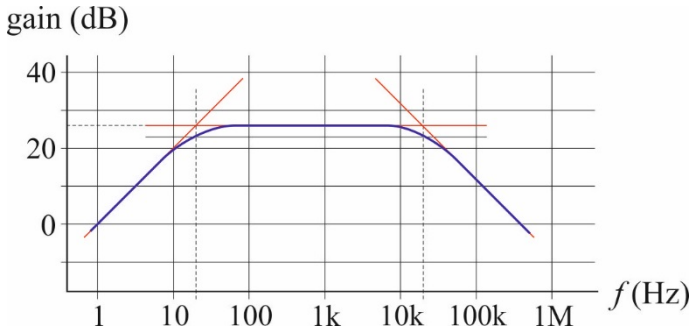
The opamp stage is a standard inverting amplifier, so the specification will be met by choosing R_1 = 1 kΩ and R_2 = 20 kΩ.

   b) If the opamp stage is required to have a band-pass characteristic with lower and upper corner frequencies of 20 Hz and 20 kHz, choose values for C_1 and C_2. Draw a dimensioned sketch showing the magnitude frequency response of the opamp stage in this case.

[10]

The lower and upper corner frequencies will be given by $f_1 = \frac{1}{2\pi R_1 C_1}$ and $f_2 = \frac{1}{2\pi R_2 C_2}$, so to meet the specification we require:
$C_1 = 1/(2\pi \times 10^3 \times 20) = 8$ µF and $C_2 = 1/(2\pi \times 2 \times 10^4 \times 2 \times 10^4) = 400$ pF.

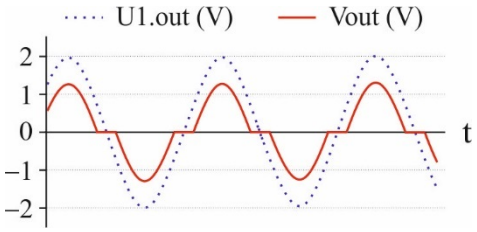
The gain in the mid-band is –20, corresponding to 26 dB, and the HP and LP filters are 1st order so the response will roll off at ±20 dB/decade. The magnitude response is therefore as shown below.



   c) The amplifier is tested and the audio quality is found to be poor. Explain why this is the case. Illustrate your answer by sketching the output waveform you would expect to see when a 1 kHz sine wave of amplitude 200 mV peak-to-peak is applied at the input.

[8]

The audio quality is poor because the Class-B output stage suffers from cross-over distortion. When $U_1.out > +0.7$, Q_1 is conducting and V_{out} is ~0.7 V lower than $U_1.out$. When $U_1.out < -0.7$, Q_2 is conducting and V_{out} is ~0.7 V higher than $U_1.out$. When $|U_1.out| < 0.7$ both output transistors are off and the output voltage is zero. With a 200 mVpp sine wave input, the opamp output will be a 4 Vpp sine wave and the outuput voltage will be as shown below.

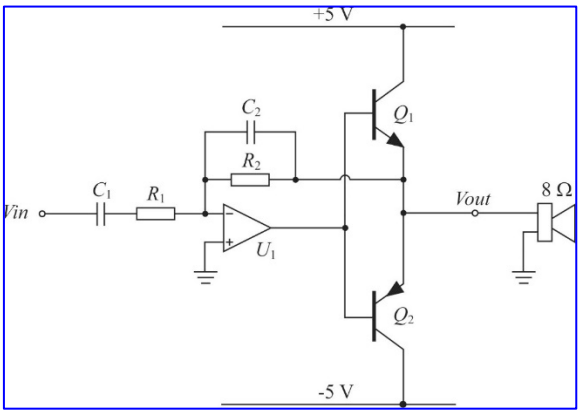


d) Sketch an alternative output stage configuration that would give improved audio quality. Can you also see a way to improve the audio quality that does not involve adding any components?

[6]

Any Class AB or Class A is acceptable as the alternative configuration. [3 marks]

The modification not requiring any additional components is to put the output stage inside the feedback loop as shown below. [3 marks]



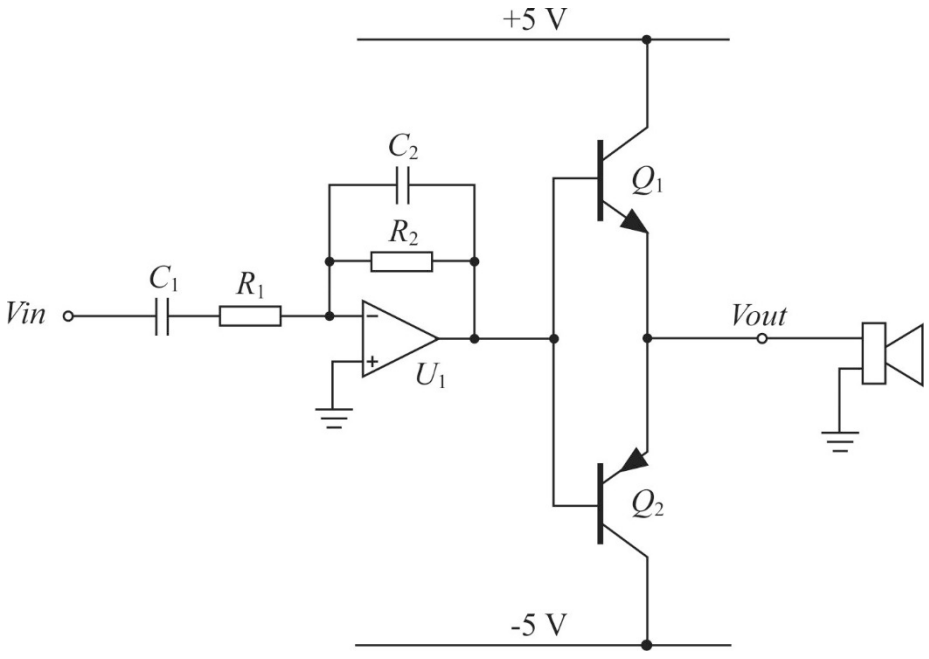


Figure 2.1

3. Figure 3.1 shows a common emitter amplifier with an active load.

a) Ignoring base currents, calculate the value of R_2 that will cause the transconductance of Q_1 to be 40 mS.

[2]

To get g_m = 40 mS, we need a collector current of $I_C = g_m V_T = 0.04 \times 0.025 = 1$ mA.
Ignoring base currents, the current in R_2 will also be 1 mA, so $R_2 = \frac{10-9.3}{1\text{m}} = 9.3$ kΩ.

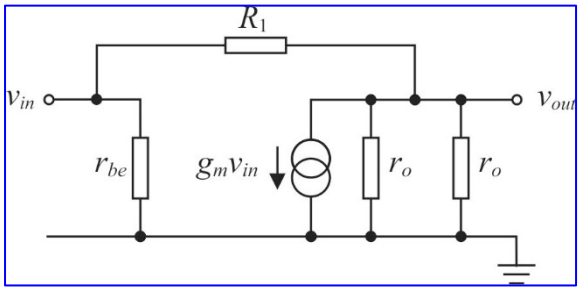
b) Calculate the value of R_1 that will bias the collector voltage of Q_1 to achieve equal head and leg room.

[2]

For equal head and leg room we need V_{out} to be 10/2 = 5 V.
The base current of Q_1 is $I_{B1} = I_{C1}/\beta$, so we have: $\frac{5-0.7}{R_1} = \frac{I_{C_1}}{\beta} = \frac{1\text{m}}{150} \Rightarrow R_1 = 645$ kΩ

c) Draw a small-signal equivalent model of the amplifier and calculate the values of any small-signal parameters. (Hint: Q_2 and Q_3 only contribute one resistor to the SSEM.)

[6]



g_m = 40 mS (from part a))

$r_{be} = \beta/g_m = 3.75$ kΩ

$r_o = V_A/I_C = 150$ kΩ

d) Calculate the small-signal input resistance R_i, output resistance R_o and voltage gain A_v of the amplifier in the mid-band.

[6]

R_o:- With the input grounded the impedance seen at the output is $R_o = r_o // r_o // R_1 = 67.2$ k.

A_v:- KCL at output: $\frac{v_{out}-v_{in}}{R_1} + g_m v_{in} + \frac{v_{out}}{\frac{r_o}{2}} = 0 \Rightarrow$

$$A_v = \frac{v_{out}}{v_{in}} = \frac{\frac{1}{R_1} - g_m}{\frac{1}{R_1} + \frac{2}{r_o}} = \left(\frac{1}{R_1} - g_m\right) R_0 = \left(\frac{1}{645k} - 0.04\right) \times 67.2k = -2688$$

R_i:- KCL at input: $i_{in} = \frac{v_{in}}{r_{be}} + \frac{v_{in}-v_{out}}{R_1} \Rightarrow$

$$R_i = \frac{v_{in}}{i_{in}} = r_{be} \parallel \left(\frac{R_1}{1 - A_v}\right) = 3.75\text{k} \parallel \left(\frac{645k}{1 + 2688}\right) = 3.75k \parallel 240 = 226\ \Omega$$

e) i) Calculate the corner frequency of the high-pass filter associated with C_{in} when the amplifier is driven by a source with resistance 50 Ω.

[3]

Input corner frequency is $f_{c_in} = (2\pi \times (50 + 226) \times 500n)^{-1} = 1.15$ kHz

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ii) Calculate the corner frequency of the high-pass filter associated with C_{out} when the amplifier is driving a 50 Ω load.

[3]

Output corner frequency is $f_{c_out} = (2\pi \times (50 + 67.2k) \times 100n)^{-1} = 23.7$ Hz

f) A 10 pF capacitor, C_1, is placed in parallel with R_1. Calculate the apparent input capacitance C_1' between the base and ground due to this capacitance. You should use the A_v value you calculated in part d) in this calculation.

[4]

The current in C_1 is $sC_1(v_{in} - v_{out}) = sC_1(1 - A_v)v_{in}$. The apparent capacitance to ground is therefore $C_1' = C_1(1 - A_v) = 10p \times 2689 = 26.9\,nF$.

g) By considering the network formed by R_i, C_1' and the source resistance, calculate the corner frequency of the low-pass filter associated with C_1 when the amplifier is driven by a source with resistance 50 Ω. You should assume that C_{in} has zero impedance for this calculation.

[4]

The filter comprises a series input resistor $R_S = 50\,\Omega$, and a shunt impedance at the output formed by R_i and C_1' in parallel. The corner frequency is:

$$f_{c\,LP} = [2\pi(R_S \parallel R_i)C_1']^{-1} = [2\pi \times (50 \parallel 226) \times 26.9n]^{-1} = 145 \text{ kHz}$$

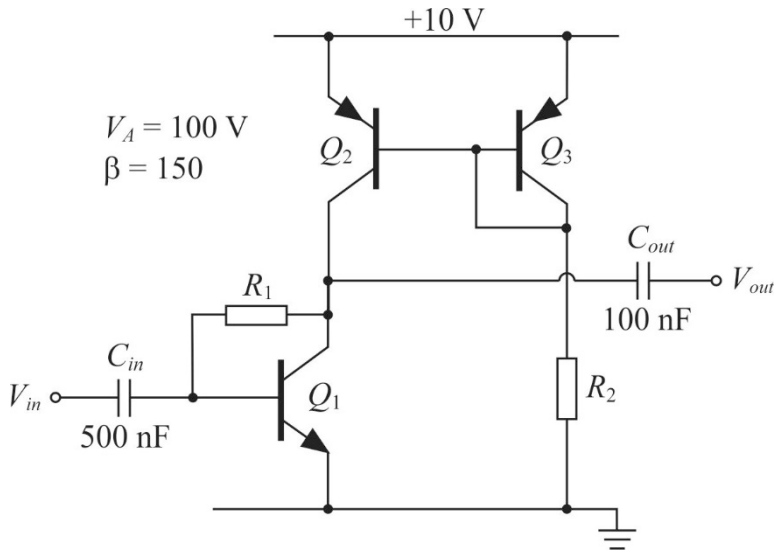


Figure 3.1